Activity 3

Take some powdered iron filings and mix it with sulphur.

i. Divide the mixture into two equal halves.

ii. Keep the first half of the mixture as it is, but heat the second half of the mixture.

iii. On heating you will get a black brittle compound.



Mixture of iron and Sulphur



Iron Sulphide compound

The black compound is Iron (II) sulphide.

$$\text{Iron + sulphur} \xrightarrow{\text{heat}} \text{Iron sulphide}$$

The Iron sulphide formed has totally different properties to the mixture of iron and sulphur as tabulated below:

Substance	Appearance	Effect of magnet
Iron (element)	Dark grey powder	Attracted to it
Sulphur (element)	Yellow powder	None
Iron + Sulphur (Mixture)	Dirty yellow powder	Iron powder attracted to it
Iron sulphide (compound)	Black solid	No effect

From the above experiment, we can summarise the major differences between mixtures and compounds:

DO YOU KNOW?

Blood is not a pure substance. It is a mixture of various components such as platelets, red and white blood corpuscles and plasma.

Table 10.2 Difference between mixtures and compounds.

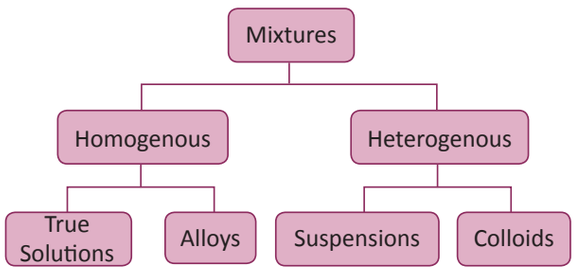
Mixture	Compound
It contains two or more substances	It is a single substance
The constituent may be present in any proportion.	The constituents are present in definite proportions.
They show the properties of their constituents.	They do not show the properties of the constituent elements.
The components may be separated easily by physical methods.	The constituents can only be separated by one or more chemical reactions.

Activity 4

Identify whether the given substance is mixture or compound and justify your answer. 1. Sand and water 2. Sand and iron filings 3. Concrete 4. Water and oil 5. Salad 6. Water 7. Carbon dioxide 8. Cement 9. Alcohol.

10.2 Types of Mixtures

Most of the substances that we use in our daily life are mixtures. In some we will be able to see the components with our naked eyes but in most others the different components are not visible. Based on this mixture can be classified as below.



10.2.1 Homogeneous and Heterogeneous mixture

A mixture in which the components cannot be seen separately is called a homogeneous mixture.

IX_SCI_EM_Unit-10_CHE.indd 116 06-02-2020 11:47:32